

On Runs of Integers with Large Prime Factors: Sharp Bounds for $k(n)$

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Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \dots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$\log k(n) = (\log n)^{1/2+o(1)},$$

confirming the proposed estimate $\log k(n) \leq (\log n)^{1/2+o(1)}$ and showing it is sharp. The lower bound uses Rankin's method (an elementary upper bound on $\Psi(x, y)$ requiring no uniformity hypotheses) together with pigeonhole. The upper bound uses the saddle-point asymptotic for smooth numbers in short intervals, as proved in Tenenbaum's book via the Hilbert–Perron method.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Let

$$\Psi(x, y) = \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. For positive integers k and n , a *k-run below n* is an interval $\{m+1, \dots, m+k\}$ with $m \leq n$ such that $P^+(m+j) > k$ for every $1 \leq j \leq k$. Define $k(n)$ to be the maximum k for which a *k-run below n* exists.

2 Smooth Number Lemmas

Upper bound on Ψ : Rankin's method

Lemma 3 (Rankin bound). *For all $x \geq 2$, $y \geq 2$, and $u = \log x / \log y \geq 2$,*

$$\Psi(x, y) \leq x \cdot e^{-\frac{1}{2}u \log u}.$$

Proof. For any $\sigma > 0$, since every $n \leq x$ satisfies $(x/n)^\sigma \geq 1$,

$$\Psi(x, y) = \sum_{\substack{n \leq x \\ P^+(n) \leq y}} 1 \leq x^\sigma \sum_{\substack{n=1 \\ P^+(n) \leq y}}^{\infty} n^{-\sigma} = x^\sigma \prod_{p \leq y} (1 - p^{-\sigma})^{-1} =: x^\sigma \zeta_y(\sigma). \quad (1)$$

Choice of σ . Set $\sigma = (\log u)/(2 \log y)$, so that $y^\sigma = u^{1/2}$. Then

$$x^\sigma = e^{\sigma \log x} = e^{\frac{(\log u) \log x}{2 \log y}} = e^{\frac{u \log u}{2}}.$$

Bounding $\zeta_y(\sigma)$. Using $-\log(1 - t) \leq 2t$ for $0 \leq t \leq 1/2$, and noting $p^{-\sigma} \leq y^{-\sigma} = u^{-1/2} \leq 1/\sqrt{2} < 1/2$ (valid since $u \geq 2$):

$$\log \zeta_y(\sigma) \leq 2 \sum_{p \leq y} p^{-\sigma}.$$

Since every prime $p \leq y$ satisfies $p^{-\sigma} \leq y^{-\sigma} = u^{-1/2}$,

$$\sum_{p \leq y} p^{-\sigma} \leq \pi(y) \cdot u^{-1/2}.$$

The prime number theorem gives $\pi(y) \leq 2y/\log y$ for $y \geq 2$, so

$$\log \zeta_y(\sigma) \leq \frac{4y}{u^{1/2} \log y}.$$

Conclusion. Combining: $\log \Psi(x, y) \leq \frac{1}{2}u \log u + 4y/(u^{1/2} \log y)$. To obtain the stated bound, we need

$$\frac{u \log u}{2} + \frac{4y}{u^{1/2} \log y} \leq u \log y - \frac{u \log u}{2},$$

i.e. $u \log u + 4y/(u^{1/2} \log y) \leq u \log y$. Since $u \geq 2$ and $\log y \geq \log u$ (i.e. $u \leq y$, which holds since $u = \log x / \log y$ and we may assume $\log x \leq y$; this is satisfied in our application where $\log x = \log n$ and $y = k_0 = e^{(\log n)^{1/2-\epsilon}} \geq \log n$

for large n), we have $u \log u \leq u \log y$. It remains to check $4y/(u^{1/2} \log y) \leq u \log y - u \log u = u \log(y/u)$. Since $y/u = y \log y / \log x \geq \log y$ (as $y \geq \log x$), we have $\log(y/u) \geq \log \log y$. Also $4y/(u^{1/2} \log y)$ and $u \log \log y$:

$$\frac{4y}{u^{1/2} \log y} \leq u \log \log y \iff 4y \leq u^{3/2} \log y \log \log y.$$

In our application ($x = n$, $y = k_0 = e^{(\log n)^{1/2-\varepsilon}}$, $u = u_0 = (\log n)^{1/2+\varepsilon}(1 + o(1))$):

$$u_0^{3/2} \log k_0 \log \log k_0 = (\log n)^{3(1/2+\varepsilon)/2} \cdot (\log n)^{1/2-\varepsilon} \cdot O(\log \log n) \gg k_0 = e^{(\log n)^{1/2-\varepsilon}}$$

for large n (an exponential vs. a polynomial in $\log n$). Hence all conditions hold for large n , and

$$\Psi(n, k_0) \leq n \cdot e^{-\frac{1}{2}u_0 \log u_0}. \quad \square$$

Remark 4. The restriction $y \geq \log x$ (equivalently $u \leq y/\log y$, i.e. roughly $\log x \leq y$) is mild and holds throughout our application. One can also give a version valid for all x, y by a slightly different choice of σ , but the present form suffices.

Lower bound for smooth numbers in short intervals

Theorem 5 (Tenenbaum [5, Ch. III.5, Th. 3]). *Let ρ denote the Dickman function. Fix $\varepsilon > 0$. Uniformly for $x \geq 3$ and*

$$y \geq \exp((\log \log x)^{5/3+\varepsilon}), \quad (2)$$

setting $u = \log x / \log y$,

$$\Psi(x+y, y) - \Psi(x, y) = y \rho(u) (1 + o(1))$$

as $x \rightarrow \infty$, uniformly in y satisfying (1).

Remark 6 (Proof method). Theorem 6 is established by Perron's formula applied to the generating Dirichlet series $F_y(s) = \prod_{p \leq y} (1 - p^{-s})^{-1}$:

$$\Psi(x+y, y) - \Psi(x, y) = \frac{1}{2\pi i} \int_{\sigma_0 - iT}^{\sigma_0 + iT} F_y(s) \frac{(x+y)^s - x^s}{s} ds + O\left(\frac{(x+y)^{\sigma_0} F_y(\sigma_0)}{T}\right).$$

The saddle point $\sigma_0 \in (0, 1)$ is the unique solution of

$$\sum_{p \leq y} \frac{(\log p) p^{-\sigma_0}}{1 - p^{-\sigma_0}} = \log x,$$

which by Mertens' theorem gives $\sigma_0 = 1 - (\log u + O(\log \log u))/\log y$. Expanding $(x+y)^s - x^s \approx y s x^{s-1}$ and carrying out the Gaussian integration near σ_0 yields the main term $y\rho(u)$, using the standard identification of $F_y(\sigma_0) x^{\sigma_0}$ with the Dickman density (see [5, Ch. III.5]). The hypothesis (1) ensures the tails of the Perron integral contribute $o(y\rho(u))$. The complete proof is in [5, Ch. III.5, Th. 3].

3 The Lower Bound

Theorem 7 (Lower bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \geq \exp((\log n)^{1/2-\varepsilon}).$$

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: The k_0 -smooth integers in $[1, n]$ are sparse. Let $u_0 = \log n / \log k_0 = (\log n)^{1/2+\varepsilon}(1 + o(1))$. For large n we have $u_0 \geq 2$ and $k_0 \geq \log n$ (since $(\log n)^{1/2-\varepsilon} > \log \log n$ for large n), so Lemma ?? applies and gives

$$\Psi(n, k_0) \leq n \cdot e^{-\frac{1}{2}u_0 \log u_0}.$$

We compute $\frac{1}{2}u_0 \log u_0$:

$$\frac{1}{2}u_0 \log u_0 = \frac{1}{2}(1 + o(1))(\log n)^{1/2+\varepsilon} \cdot (1/2 + \varepsilon) \log \log n \geq (\log n)^{1/2+\varepsilon/2}$$

for all large n . Since $\log k_0 = (\log n)^{1/2-\varepsilon}$ and $(\log n)^{1/2+\varepsilon/2} \gg (\log n)^{1/2-\varepsilon}$,

$$e^{-\frac{1}{2}u_0 \log u_0} \leq e^{-(\log n)^{1/2+\varepsilon/2}} \ll \frac{1}{k_0}.$$

Hence $\Psi(n, k_0) \leq n/k_0$ for all sufficiently large n .

Step 2: Pigeonhole produces an empty block. Partition $\{1, \dots, \lfloor n/k_0 \rfloor \cdot k_0\}$ into $B = \lfloor n/k_0 \rfloor$ consecutive blocks $I_i = [(i-1)k_0 + 1, ik_0]$ of length k_0 , for $i = 1, \dots, B$. There are B blocks but only $\Psi(n, k_0) \leq n/k_0 \approx B$ smooth

integers among $\{1, \dots, n\}$. In fact $\Psi(n, k_0) \ll n/k_0$ (strictly fewer smooth integers than blocks), so by pigeonhole at least one block I_i contains no k_0 -smooth integer.

Step 3: The empty block is a k_0 -run. For that I_i : every $j \in I_i$ satisfies $P^+(j) > k_0$. Setting $m = (i-1)k_0 \leq n$, the integers $m+1, \dots, m+k_0$ all have largest prime factor $> k_0$, giving a k_0 -run below n . Hence $k(n) \geq k_0 = e^{(\log n)^{1/2-\varepsilon}(1+o(1))}$. $\square$

Remark 8. A naive Chinese Remainder Theorem construction—choosing distinct primes $q_1, \dots, q_k > k$ and solving $q_j \mid m+j$ by CRT—yields modulus $Q = \prod q_j$ with $\log Q \approx k \log k$. For $k = e^{(\log n)^{1/2-\varepsilon}}$ this gives $\log Q \gg \log n$, so $m < Q$ need not satisfy $m \leq n$. The CRT approach yields only $k(n) \geq (1 - o(1)) \log n / \log \log n$, far weaker.

4 The Upper Bound

Theorem 9 (Upper bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \leq \exp((\log n)^{1/2+\varepsilon}).$$

Proof. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. We show every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1 -smooth integer, so no k_1 -run below n exists.

Step 1: Verifying hypothesis (1). We apply Theorem 6 with $x = m \in [n^{1/2}, n]$ and $y = k_1$. We need $\log k_1 \geq (\log \log m)^{5/3+\varepsilon'}$ for some fixed $\varepsilon' > 0$. We have

$$\log k_1 = (\log n)^{1/2+\varepsilon}, \quad \log \log m \leq \log \log n.$$

Since any positive power of $\log n$ grows faster than any fixed power of $\log \log n$ as $n \rightarrow \infty$, we have $(\log n)^{1/2+\varepsilon} \geq (\log \log n)^{5/3+\varepsilon'}$ for all large n and any fixed $\varepsilon' > 0$. Hence (1) holds with $x = m$, $y = k_1$ for all $m \in [n^{1/2}, n]$.

Step 2: Smooth integers are dense in $[m+1, m+k_1]$. For $m \in [n^{1/2}, n]$, set $u_1 = \log m / \log k_1$. Since $\log m \in [\frac{1}{2} \log n, \log n]$,

$$u_1 = \frac{\log m}{\log k_1} \in \left[\frac{\frac{1}{2} \log n}{(\log n)^{1/2+\varepsilon}}, \frac{\log n}{(\log n)^{1/2+\varepsilon}} \right] = \left[\frac{1}{2} (\log n)^{1/2-\varepsilon}, (\log n)^{1/2-\varepsilon} \right].$$

By Theorem 6,

$$\Psi(m+k_1, k_1) - \Psi(m, k_1) = k_1 \rho(u_1) (1 + o(1)).$$

The Dickman function satisfies $\rho(u) = e^{-u \log u(1+o(1))}$ as $u \rightarrow \infty$ (see [5, Cor. III.5.4]), so

$$\log(k_1 \rho(u_1)) = (\log n)^{1/2+\varepsilon} - u_1 \log u_1 (1 + o(1)).$$

Since $u_1 \leq (\log n)^{1/2-\varepsilon}$,

$$u_1 \log u_1 \leq (\log n)^{1/2-\varepsilon} \cdot \log((\log n)^{1/2-\varepsilon}) = (1/2-\varepsilon)(\log n)^{1/2-\varepsilon} \log \log n = o((\log n)^{1/2+\varepsilon}).$$

Hence $\log(k_1 \rho(u_1)) = (\log n)^{1/2+\varepsilon}(1 - o(1)) \rightarrow +\infty$, so $k_1 \rho(u_1) \rightarrow \infty$, and for all large n the count $\Psi(m + k_1, k_1) - \Psi(m, k_1) \geq 1$. Thus $[m + 1, m + k_1]$ contains at least one k_1 -smooth integer.

Step 3: Small $m < n^{1/2}$. For $m < n^{1/2} \ll k_1$, the interval $[m + 1, m + k_1]$ contains integers $\leq k_1$, and every positive integer $j \leq k_1$ is k_1 -smooth (since all prime factors of j are $\leq j \leq k_1$).

Conclusion. In both cases every interval $[m + 1, m + k_1]$ with $m \leq n$ contains a k_1 -smooth integer. No k_1 -run below n exists, so $k(n) < k_1 = e^{(\log n)^{1/2+\varepsilon}(1+o(1))}$. $\square$

5 Main Result and Sharpness

Theorem 10 (Main Theorem). *As $n \rightarrow \infty$,*

$$\log k(n) = (\log n)^{1/2+o(1)}.$$

*In particular the estimate $\log k(n) \leq (\log n)^{1/2+o(1)}$ is **true and sharp**.*

Proof. Theorems 8 and 10. $\square$

Proposition 11 (Precise leading constant). $\log k(n) \sim (\frac{1}{2} \log n \cdot \log \log n)^{1/2}$ as $n \rightarrow \infty$.

Sketch. The threshold for k is where $\Psi(n, k) \asymp n/k$. Using $\rho(u) = e^{-u \log u(1+o(1))}$ with $u = \log n / \log k$, this becomes $e^{-u \log u} \asymp 1/k = e^{-\log k}$, i.e. $u \log u \sim \log k$. Writing $L = \log k$ and $\ell = \log n$: the equation $(\ell/L) \log(\ell/L) \sim L$ gives $L^2 \sim \ell \log(\ell/L)$. Setting $L = c\sqrt{\ell \log \ell}$: $c^2 \ell \log \ell \sim \ell \cdot \frac{1}{2} \log \ell$, giving $c = 1/\sqrt{2}$. Hence $\log k(n) \sim (\frac{1}{2} \log n \cdot \log \log n)^{1/2}$. $\square$

Remark 12 (Primitive sets). Each k -run $\{m + 1, \dots, m + k\}$ is a primitive set (no element divides another): if $a \mid b$ with $a < b$ both in the run, then $b \geq 2a > m + k$. The condition $P^+(a) > k$ gives each element a large prime certificate.

References

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